Strathmore UNIVERSITY

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LAB SHEET

DBT 1305

Four Pages

The Ereteti site, marked up by hand and validating

WEEK	Week 2 · 14 – 17 September
SESSION	Session 2 of 2 — Lab, Thursday
MODE	Laboratory practical
DURATION	2 hours
LEARNING OUTCOMES	ILO1 · ILO2
ACADEMIC SESSION	Year 1, Semester 3

WEEK 2 | LAB · NOT MARKED | APPLY | 2 HOURS

PURPOSE

Build the four public pages of the Ereteti Dairy Sacco site — `index.html`, `about.html`, `routes.html` and `contact.html` — linked to each other, marked up with semantic HTML5, every page passing the W3C validator, and all of it pushed to GitHub before you stand up.

This lab is not marked. **Lab 1, next Thursday, is** — and it is built directly on top of what you finish today. A student who leaves today with two pages starts next week behind.

You write these by hand. Do not paste them from anywhere, and do not use an extension that writes the tags for you. This is the only week where the typing is genuinely the point: you are learning the shape of the language, and you learn it through your fingers.

NOTE

The words are given to you. The tags are not. A separate sheet — `page-text-W2.txt` — carries the sentences for all four pages, and three photographs are attached beside it on the **Lab Activity** post in Classroom. Copy the words from it and save yourself inventing paragraphs about a sacco you have never visited. **Every angle bracket on every page is still typed by you**, and there is not a single tag in that file to copy even if you wanted to.

WATCH OUT

The code on this sheet has gaps in it. Every `_____` is a missing tag name, attribute or closing tag that you work out and type yourself. There are nine of them, they are all short, and every one is answerable from the notes or from the lines around it. Copy a block without filling its gap and the validator in Part 4 will fail the page.

They are there on purpose, and the purpose is your hands. A page you pasted teaches you nothing; a page you typed leaves you knowing, without looking, that every `<ul>` wants a `</ul>` and that an `img` is nothing without its `alt`. Nobody reaches that by reading. The gaps are small so the lab still finishes on time — they cost you about a minute in total, and they are the minute the rest of the semester is built on.

WATCH OUT

No CSS today. None. No stylesheet, no `<style>` block, no `style="..."` attribute. Your pages will be black text on a white background with bullets down the left, and that is exactly right — it is the only week you get to see what markup alone does. Week 4 changes all of it, from one file.

Part 0 — If last Thursday did not work

Skip this if your page opened through Live Server and your repository is on GitHub. If not, fix it now — the rest of the sheet needs it.

What you see	What to do
The page shows your code instead of a heading	The file saved as `index.html.txt`. Rename it. In Windows Explorer turn on **File name extensions** first, or you cannot see the problem
`git: command not found`	Git is installed but this terminal was opened before it was. Close the terminal, open a new one
GitHub asks for a password and refuses yours	GitHub no longer accepts account passwords. `Settings` → `Developer settings` → `Personal access tokens` → generate one, and paste that as the password
`failed to push some refs`	You created the GitHub repository with a README, so it has a commit yours does not. Run `git pull --rebase origin main`, then push again
The address bar says `file:///`	You opened the file directly. Right-click `index.html` in VS Code and choose **Open with Live Server**
`remote origin already exists`	You ran the remote step twice. `git remote remove origin`, then add it again

> **CHECKLIST**
>
> Do not go past Part 0 until all four are true.
>
> - VS Code has the folder `ereteti-sacco` open — the folder, not a file
> - `index.html` opens through Live Server at `http://127.0.0.1:5500`
> - `git log` lists at least one commit with your name on it
> - Your repository is open in a browser tab right now, and it is **public**

Part 1 — The skeleton, and the shared parts

Every one of today's four pages has the same header, the same navigation and the same footer. You write them once, here, and copy them three times.

1 — Make the folder for pictures. In the VS Code Explorer, create a folder inside `ereteti-sacco` called exactly `img` — lower-case, three letters. Leave it empty for now.

2 — Open `index.html` and replace everything in it. The page you wrote last week has done its job. Type this in:

```
<!DOCTYPE html>
<html lang="______">
<head>
  <meta charset="utf-8">
  <title>Ereteti Dairy Sacco</title>
</head>
<body>

  <header>
    <h1>Ereteti Dairy Sacco</h1>
    <p>Milk collected twice a day, on three routes out of Ngong.</p>
  </header>

  <nav>
    <ul>
      <li><a href="index.html">Home</a></li>
      <li><a href="about.html">About</a></li>
      <li><a href="routes.html">Our routes</a></li>
      <li><a href="contact.html">Contact</a></li>
    </ul>
  </______>

  <main>

  </main>

  <footer>
    <p>Ereteti Dairy Sacco, PO Box 4021, Ngong.</p>
    <p>Site built by Wanjiru Kamau, DBT Year 1, 2026.</p>
  </footer>

</body>
</html>
```

Put your own name in the footer, not Wanjiru's. `main` is deliberately empty — you fill it in Part 2.

3 — Save it and look at it. Through Live Server. You should see the heading, four bullets that are links, and the footer. Click one of the links: it will fail, because the other three pages do not exist yet. That is expected, and Part 3 fixes it.

4 — Read your own navigation. Each `li` holds one `a`. The list is unordered because no page comes before another. The links are relative — no slash at the front — so they look for a file next to this one.

> **NOTE**
>
> **Why is the navigation a bulleted list going down the page?** Because there is no CSS. A `ul` is drawn with bullets, one item per line, until something says otherwise. In Week 4 four lines of CSS turn this into a bar across the top, and the HTML does not change at all. That is the whole argument for marking up meaning rather than appearance.

Part 2 — The home page

Everything from here goes **inside** `main`, between the tags you left empty.

1 — Write the first section. A heading, and two paragraphs — what the sacco is, and who may join. The sentences are on the text sheet under **PAGE 1 OF 4**; copy them from there rather than typing placeholder text.

```
<_____>
  <h2>Milk collected twice a day, paid once a month</h2>
  <p>Ereteti Dairy Sacco has collected milk on the Ngong road since
  1998. Around eight hundred farmers deliver to us twice a day, and
  we pay for every accepted litre once a month.</p>
  <p>Any farmer on one of our three routes may join. You need a
  national ID and a bank account, and you may deliver from the day
  your membership number is issued.</p>
</section>
```

2 — Write the second section, with a list. The three routes, each one a link through to `routes.html`:

```
<section>
  <h2>Where we collect</h2>
  <ul>
    <li><a href="_____">Kiserian route</a></li>
    <li><a href="routes.html">Ngong route</a></li>
    <li><a href="routes.html">Rongai route</a></li>
  </ul>
</section>
```

3 — Add one image. Put a photograph into the `img` folder first. **Three are attached to the Lab Activity post in Classroom** — download `collection-centre.jpg` into your `img` folder and use that one here. They are already sized correctly. A picture you took yourself is a better answer if you have one; resize it to about 1200 pixels wide before you use it.

```
  <img src="img/collection-centre.jpg"
       _____="Farmers queuing with churns at the Kiserian centre">
```

Use the real file name, and make the `alt` text say what is actually in your picture. "image" is not alt text.

4 — Check your heading levels. You now have one `h1` in the header and two `h2` inside `main`. That is an outline: the page, and two sections of it. No `h3` yet, and no second `h1` anywhere.

5 — Save, look, and commit.

```
git add .
git commit -m "Week 2 — home page marked up"
```

Part 3 — The other three pages

Three times the same move: copy `index.html`, change the title, change what is inside `main`. The header, nav and footer stay identical.

1 — Make `about.html`. Copy `index.html`, rename the copy, then:

- Change the `title` to `About us — Ereteti Dairy Sacco`
- Change the `h1` in the header to `About Ereteti`
- Replace the contents of `main` with: an `h2`, two paragraphs of history, one image with real `alt` text, and a `blockquote` holding one sentence from the chairperson — all of it under **PAGE 2 OF 4** on the text sheet, including which words earn the `strong` and the `em`

```
<blockquote>
  <______>We started with nine farmers and one cooler. The ledger was a
  school exercise book.</______>
</blockquote>
```

Use `<strong>` and `<em>` at least once each somewhere on this page — and only where the meaning actually calls for them.

2 — Make `routes.html`. Copy `index.html` again. Title it `Our routes — Ereteti Dairy Sacco`, and put **three sections** inside `main`, one per route. Each section has the same shape:

```
<section>
  <h2>Kiserian route</h2>
  <p>The lorry runs from 6.30am and again from 4.00pm.</p>
  <ul>
    <li>Olepolos centre</li>
    <li>Kandisi centre</li>
  </______>
</section>
```

The centres and times for all three routes are on the text sheet under **PAGE 3 OF 4** — three sections, same shape, different words. In Week 10 this page stops being typed out by hand and starts coming from the database instead — the markup you write today is what the PHP fills in.

3 — Make `contact.html`. Title it `Contact us — Ereteti Dairy Sacco`, and inside `main`:

- An `h2` and the office address, in an `address` element
- An `h2` and a list of two telephone numbers, each one a `tel:` link
- A `mailto:` link for the secretary

```
      <address>
        Ereteti Dairy Sacco<br>
        PO Box 4021<br>
        Ngong, Kajiado County
      </address>

      <ul>
        <li><a href="tel:_____">0720 000 000</a> — the office</li>
        <li><a href="tel:+254733000000">0733 000 000</a> — the Kiserian centre</li>
      </ul>
```

There is **no contact form** on this page. Forms are Week 3.

4 — Click every link on every page. Four pages, four links each, sixteen clicks. Every one of them must land somewhere. This is the step everybody skips and it is the one the marker checks first.

CHECKLIST

- No `_____` is left anywhere in any of the four files
- Four files exist, all lower-case, no spaces in any file name
- Each page has a **different** `title`
- Each page has exactly one `h1`, and no heading level is skipped
- The header, nav and footer are identical on all four
- Every one of the sixteen links works

Part 4 — The validator

This is the part that decides whether today worked.

1 — Go to `validator.w3.org`. Choose **Validate by Direct Input** and paste one page in. Or use **Validate by File Upload** and give it the file.

2 — Read what it says. Green, with no errors, is the target. An error tells you the line number and what it expected — read it as a sentence, because it is usually saying exactly what is wrong.

3 — Fix, and run it again. Then do the other three pages.

The five you are most likely to see today:

What it says	What it means
`End tag for element "p" which is not open`	You closed something you had not opened, or you crossed two tags
`An img element must have an alt attribute`	Exactly that. Add it, and make it say something

What it says	What it means
`Duplicate ID` or `Element h1 not allowed here`	You have two `h1` elements, or one inside something that cannot hold it
`Stray end tag`	An extra closing tag, usually left behind when you copied a block
`Unclosed element "ul"`	A missing `</ul>` or `</li>` — look at your indentation and the mistake is usually visible
`Element name "_____" not allowed` or similar	You copied a block off this sheet and left one of its nine gaps unfilled

NOTE

Warnings are not errors. The validator may warn about a section with no heading, or about the `lang` attribute. Green with warnings is a pass today. Red is not.

Part 5 — Push it

```
git add .
git commit -m "Week 2 — four pages, all validating"
git push
```

Then **reload your repository page on GitHub** and check that all four `.html` files and the `img` folder are listed. If they are not there, they are not pushed, and nothing on your laptop counts.

CHECKLIST

You may leave once all of these are true. Show the demonstrator your screen.

- Four pages, each opening through Live Server
- Every gap filled — search each file for `_____` and find nothing
- All sixteen navigation links work
- All four pages validate with no errors
- No CSS anywhere — no stylesheet, no `<style>`, no `style="..."`
- All four files are visible in your GitHub repository in the browser

TASK

Before Monday. Add a second image to `about.html` with proper `alt` text, re-validate the page, and push that change. Doing it once more on your own, without this sheet in front of you, is how it stops being copying.

If something will not work

What you see	What it usually is
A link does nothing	The file name does not match the `href` — check the capitals and the spelling, one character at a time
The whole page after a certain point vanishes	A missing closing quote on an attribute. Everything after it was swallowed
The page looks fine but the validator is red	It is still wrong. The browser guesses at broken markup; the validator does not, and Week 13's server will not either
Text runs together that should be separate paragraphs	You used line breaks instead of `<p>` elements. The browser ignores your line breaks
The image does not appear	The path in `src` is wrong, or the file is not in `img/`, or the extension is `.JPG` and you typed `.jpg`
Everything is bold from halfway down	An unclosed `<strong>`. The validator will name the line
Nothing above matches	Ask. Do not spend forty minutes on it — today is about the markup, not about fighting the tools

What happens to these pages

Nothing you wrote today is thrown away.

Week	What happens to it
4	`css/style.css` — the sacco's colours and type, applied to all four
6	Laid out with Grid and Flexbox, and made to work at phone width
9	The header and footer become `includes/header.php` and `includes/footer.php`
10	`routes.html` becomes `routes.php` and its routes come from MySQL
13	All of it goes live, on a real host and a real domain